A

PROJECT REPORT

ON

EXCELLENCE OF VIGENERE CIPHER

Submitted to

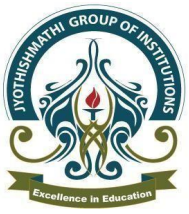
STATE BOARD OF TECHNICAL EDUCATION & TRAINING, TS

In partial fulfillment of the requirement for the award of the degree of

DIPLOMA
In
COMPUTER ENGINEERING
Submitted by

V.AARUSH **21261-CS-091**

Under the Esteemed guidance of
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DEPARTMENT OF COMPUTER ENGINEERING

JYOTHISHMATHI INSTITUTE OF TECHNOLOGY&SCIENCE

SECOND SHIFT POLYTECHNIC -261

[Autonomous]

(Accredited by NAAC with "A" Grade, NBA, Approved by A.I.C.T. Affiliated to SBTET, Hyderabad)

Nustulapur, Karimnagar – 505481, Telangana

(2021–2024)

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Telangana (2021–2024)

DEPARTMENT OF COMPUTER ENGINEERING

CERTIFICATE

This is to certify that **V.AARUSH 21261-CS-091** has successfully completed the project entitled "EXCELLENCE OF VIGENERE CIPHER " in the partial fulfillment for the award of degree of Diploma in computer engineering during the academic year 2023-2024.

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DECLARATION

I hereby declare that this Project Report on **"EXCELLENCE OF VIGENERE CIPHER"** submitted in the department of **Computer Engineering,** Jyothishmathi Institute of Technology and Science, Karimnagar, in partial fulfillment of the requirement for the award of the **Diploma in Computer Engineering** is a Bonafide record of my own work carried out under the supervision of **Mrs.B.JAYASRI, Assistant Professor, CSE Dept.**

Also, I declare that the matter embodied in this report has not been submitted for the award of any degree of any other institution or university previously.

Date:

Place: Karimnagar,

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ACKNOWLEDGEMENT

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I also thank all the **Faculty Members**, Department of Computer Engineering for their encouragement and assistance.

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ABSTRACT

Data security is a major issue which we are facing today in this digital world of communication. As we know that today hackers are almost at every corner in search of our useful data which can be hacked by them for their different purposes. Even the risk gets doubled when come to the data of any country's government. So, a system or terminology is must required to make that data safe forever by any means during communication.

Data protection can be accomplished by changing the original data by any means to some other unuseful data so that if someone gets that data then also it must remain in unuseful bits. This process can be achieved by Encrypting that data by some means of algorithms which are known to the sender and the similar Decryption algorithms to be known to only the desired receiver so that it can convert that encrypted data back to the user understandable form. Today as it is a need to develop such kind of applications which performs the specified task but along with it should be very much user friendly so that no special skills need to be required to learn in order to use that application or project.

In the application the user has to select either wants to send something by encrypting or wants to receive by decrypting. If it wants to send then it have to select source file previously designed or type some message which is to Encrypt and then transfer. Whereas on the receiver side again the receiver have to select the file which is to be received from the sender along with a decryption key to decrypt that message. Decryption key can be selected either manually if told by the sender or selecting key sent by the sender along with the encrypted data to avoid further delay in processing the message.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO VIGENERECIPHER:

The Vigenère Cipher, invented by the French cryptographer Blaise de Vigenère in the 16th century, is a method of encrypting and decrypting messages. It is a poly alphabetic substitution cipher, which means that it uses multiple cipher alphabets to encrypt the plain text.

The Vigenère Cipher operates by using a keyword, typically a word or phrase, as the basis for encryption. The keyword is repeated to match the length of the plain text message. Each letter in the keyword is then used to determine the shift value for the corresponding letter in the plain text.

To encrypt a message, each letter of the plain text is shifted by the corresponding letter in the keyword. This is done using a table called the Vigenère square or Vigenère tableau. The Vigenère square is a grid of alphabets, where each row represents a shift of the previous row by one position. The letter at the intersection of the keyword letter and plain text letter in the Vigenère square gives the encrypted letter.

For example, let's encrypt the word "HELLO" using the keyword "KEY". First, repeat the keyword to match the length of the plain text: "KEYKE". Then, find the corresponding letters in the Vigenère square. The letter 'H' is shifted by 'K', resulting in 'S'. 'E' is shifted by 'E', resulting in 'X'. 'L' is shifted by 'Y', resulting in 'M'. And so on. The encrypted message would be "SXMMR".

Decryption works in a similar way. Each letter of the cipher text is shifted back by the corresponding letter in the keyword, using the Vigenère square. By reversing the process, the original plain text message can be obtained.

The Vigenère Cipher offers more security compared to simple substitution ciphers because it introduces variability and makes it harder to identify patterns in the encrypted message. A square cipher table formed by placing the same normal or mixed primary alphabet one step farther to the left on each successive line and used by reading the cipher text letter within the table in the row and column defined by key and plain text letters in alphabets in the left and top margins respectively

1.2 PROJECT DEFINITION:

In this project, we propose to develop a system to hiding data by using “VIGENERE CIPHER” technique as I used many methods stands on some techniques to have at the back-end a software for hiding data based on hiding algorithms.

After studying the data hiding algorithms we found many ways to hiding data by using the multimedia files and the main question for me was “Where hidden data hides?” as we found by our search to know where the data hides it’s important to know what is file type of the data that is shall be hidden and the cover file type so it is possible to alter graphic or text files slightly without losing their overall viability for the viewer and listener.

By the final of our Research, we developed a Graphical User Interface uses an algorithm, to embed data in an text. The purposed system is called “Vigenère Cipher”, the aim of this project is to encrypt the data, the meaning of encrypt is to hide the data over an text using different Vigenere Cipher algorithms, in this system LSB is the algorithm that we use to hiding the data.

The key concern I had after studying the data concealing techniques was, "Where hidden data hides?" We discovered numerous methods of data hiding using data or text files. As we discovered via our research, it's crucial to understand the file type of the data that will be hidden as well as the cover file type in order to make minor changes to graphic or text files without compromising their overall suitability for the viewer or listener.

By the end of our research, we created a graphical user interface that uses an algorithm to embed data in an secret text. This system, which is called " Encryption And Decryption Using Vigenere Cipher" was created with the goal of encrypting the data.Vigenere Cipher algorithm that we use in this system to hide the data.

1.3 STATEMENT OF PROBLEM:

This project addresses the security problem of transmitting the data over internet network, the main idea coming when we start asking that how can we send a message secretly to the destination? The science of Vigenère cipher answers this question. Using Vigenère cipher information can be hidden in carriers such as text files and data transmissions.

The main purpose of this project is to secure the data so that it can be transferred through the internet or any other communication sources. As we are in the peak age of technological advancement internet usage has become inseparable part of our life.

Communication and data transfer has become a common habit of ours these days. But with all these facilities there are risks too. People on internet now haw abilities to steal data from our system or during data transfer through the internet. To solve this problem we developed our program to generate a secret code using a user defined key to encrypt the data so it is hard for other people to decipher the code.

In this document, we proposed some methods and algorithms of an Vigenère cipher systems to hide a text of a secret message.

1.4 PROJECT OBJECTIVES:

In this project we primarily concentrated on the data security issues when sending the data over the network using Vigenère Cipher techniques.

The main objectives of our project are to product security tool based on Vigenère Cipher techniques to hider message carried by Vigenère Cipher which should not be sensible to human beings and avoid drawing suspicion to the existence of hidden message.

1.5 PROJECT SCOPE:

Our project scope is developed for hiding information in any text file to ensure the safety of exchange the data between different parties and provide better security during message transmission.

The scope of the project is implementation of Vigenère cipher tools for hiding information includes any type of information file and text files and the path where the user wants to save the text and extruded file. We will use LSB technique; the proposed approach is to use the suitable algorithm for embedding the data in an text file; we will show a brief of this algorithm that we used to hiding data.

1.6 VIGENERE CIPHERE ARCHITECTURE:

In the proposed system we concentrate on finding some algorithm to hide the data inside Text using Vigenere Cipher technique. An algorithm is designed to hide all the data inputted within the text protect the privacy of the data. Then, the system is developed based on the new Vigenere cipher algorithm.

The stages of the encryption process in the system architecture include:

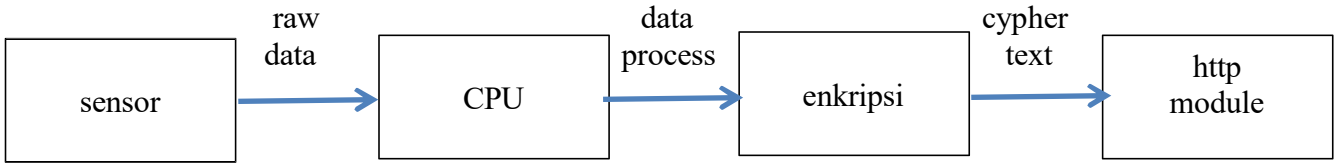
1. The client opens a connection to the Server to receive public keys (n, e)

2. The server sent a public key

3. The client receives a public key

4. The client does the plain text encryption with the public key 5. The client sends the encrypted cipher text to the server

6. The server accepts cipher text and decrypt it using the private key



Data security is needed to keep the message confidential and can only be understood by the recipient and sender. After the sensor captures information, the sensor will send raw data to the CPU and the CPU will process the received data. These data will be encrypted to ensure its safety. This encryption process will produce cipher text data and will be accepted by the server.

CHAPTER 2
LITERATURE REVIEW

In modified version of Vigenere algorithm was proposed in which diffusion is provided by adding a random bit to each byte before the message is encrypted using Vigenere. This technique fails kasiski attack to find the length of key because the padding of message with random bits. The main drawback of this technique is that the size of the encrypted message will be increased by around 56%.

In a new way of implementing Vigenere algorithm was introduced by automatically changing the cipher key after each encryption step. In this method successive keys were used that were dependent on the initial key value during the encryption process.

In modification of Vigenère cipher by random numbers, punctuation's & mathematical symbols was presented. In proposed method numbers, Punctuations and mathematical symbols were used for key in place of characters to make it more difficult for brute force attack. It was concluded that if random numbers are used for key and to spread the spectrum then only skilled persons can identify the message.

A novel approach was presented by combing the LFSR (Linear feedback shift register) key with Vigenère cipher key. In Proposed technique the concepts and methodologies of classical Vigenère cipher plus modern LFSR stream cipher was used. Proposed technique generates the period less and pseudo random letter key stream for encryption/ decryption. Period less and randomness of key lowers the flatness of letter frequency that makes it difficult to identify the length of key.

In the Caesar Cipher and Vigenère Cipher have been modified and expanded by including alphabets, numbers and symbols and at the same time introduced a complete confusion and diffusion into the modified cipher developed. It was concluded that cipher text generated by proposed hybrid technique is very difficult to break using a frequency method, brute force attack etc.

A new algorithm by combining Vigenère substitution cipher with Stream cipher was proposed in which repeated portions of plain text always encrypted with the different portion of the keyword or binary key. The letters in odd location were encrypted with stream cipher and the letters in even locations with Vigenère cipher. It was concluded that proposed algorithm hides the relation between cipher text and plain text that makes cryptanalysis much difficult.

CHAPTER 3
REQUIREMENTS

3.1 EXISTANCE OF VIGENERE CIPHER SYSTEMS:

As we know Vigenere Cipher is a method of concealing secret data inside various text like data. We are currently working on Vigenere cipher it is a process of hiding our secret data inside the text. Various Vigenere cipher systems have been developed by various programmers.In these applications mostly, there will be options like insert text, key for encryption and same process for decryption. In the encryption process text will be converted into cipher text and then using the same application the receiver can decrypt the text by using the key and convert the text into plain text.

3.2 OVERVIEW OF PROPOSED SYSTEM:

In traditional Vigenere cipher each alphabet has one fixed numeric value but in our proposed technique we have eight tables. In each table every alphabet represent with different numeric value. In traditional Vigenere technique the plain text is considered as a sequence of alphabets without any space between them. It may create a problem for receiver to read the message by inserting spaces between words and receiver needs to guess the exact place to insert space in decrypted plain text. In proposed technique we eliminate this problem by introducing different numeric value for space in each table. The encryption and decryption process by proposed approach is given below:

3.2.1 Encryption:

Formula for encryption by proposed method is: Ci = Pi + Ki (mod m) In proposed approach we have length of alphabet 27, so value m will be 27. The steps for encryption process are:

If the length of key is smaller than the length of plain text then key will be repetitive until it becomes equal to the length of plain text. Numeric value of first plain text character and key character will be added according to table 1.

Modulo 27 of the resultant value from above step will be calculated. The character correspond to the calculated modulo value will be first cipher text character. Numeric value of second plain text character and key character will be added according to table 2. Modulo 27 of the resultant value from above step will be calculated. The character correspond to the calculated modulo value will be second cipher character.

The process explained in above steps will remain continue till eighth table. After that next plain character i.e character 9 of plain text and key will undergo through same process by using value from table 1 and so on. Mathematically we can express encryption process by proposed algorithm as:

C1 = P1 + K1 (mod 27) [T1], C2 = P2 + K2 (mod 27) [T2], , C8 = P8 + K8 (mod 27) [T8], C9 = P9 + K9 (mod 27) [T1], , C16 = P16 + K16 (mod 27) [T8], C17 = P17 + K17 (mod 27) [T1],------

Where, T in above mathematical relation represents table no.

3.2.2 Decryption:

Decryption process of proposed approach works the same way as encryption does but in reverse direction. Formula for decryption by proposed method is:

Pi = Ci - Ki (mod m)

The steps for decryption process are:

Numeric value of first cipher text character and key character will be subtracted according to table 1. Modulo 27 of the resultant value from above step will be calculated. The character correspond to the calculate modulo value will be first plain text character. The process explained in above steps will remain continue till eighth table. After that next cipher character i.e character 9 of cipher text and key will undergo through same process by using value from table 1 and so on. Mathematically we can express decryption process by proposed algorithm as:

P1 = C1 - K1 (mod 27) [T1], P2 = C2 - K2 (mod 27)
[T2], ------- , P8 = C8 - K8 (mod 27) [T8], P9 = C9 - K9
(mod 27) [T1], ------, P16 = C16 - K16 (mod 27) [T8], P17
= C17 - K17 (mod 27) [T1],------

T.No.	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
1	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	&	A
2	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	&	A	B	C
3	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	&	A	B	C	D	E
4	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	&	A	B	C	D	E	F	G
5	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	&	A	B	C	D	E	F	G	H	I
6	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	&	A	B	C	D	E	F	G	H	I	J	K
7	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	&	A	B	C	D	E	F	G	H	I	J	K	L	M
8	P	Q	R	S	T	U	V	W	X	Y	Z	&	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O

Proposed technique table.

The first column represent the table numbers and first row represent the numeric value of alphabets in each table. Space is represented by —&‖ in given table.

In our proposed method space in the plain text will also converted into some cipher character by using the value of space from tables.

Example:

Let's suppose our plain text is "RED COOKIES" and key phrase is "GAME"‖.

Encryption and decryption by proposed approach

Plain Text	R	E	D	&	C	O	O	K	I	E	S
Key	G	A	M	E	G	A	M	E	G	A	M
Cipher text (encryption)	R+G (mod 27) = 21 mod 27= 21 = W [T1]	B	K	X	&	N	N	J	N	B	Z
Plain text (decryption)	W –G (mod 27)= 16 (mod 27)= 16 = R [T2]	E	D	&	C	O	O	K	I	E	S

In proposed technique each combination of plain text character and key phrase character can replaced with several cipher characters because of multiple tables but in traditional Vigenere technique there is exactly one value for each combination. Proposed approach makes our technique much stronger against kasiski and Friedman attack to find the length of key. As we can see in example that proposed technique also converted the space between the words into cipher text that will be helpful for receiver to read the plain text message easily after decryption process.

3.3 SYSTEM REQUIREMENTS:

3.3.1 FUNCTIONAL REQUIREMENTS:

Functional requirements are the requirements that define specific behavior or function of the system.

- **Secret Text Message**: The message which you want to hide should be written in this label..
- Sender send the encrypted text and generated key to receiver.
- Receiver receives the encrypted text and then decrypt the encrypted text with key.
- Either Decryption key can be selected either manually if told by sender or selecting

key sent by the sender along with the encrypted data to avoid further delay in processing the message..

3.3.2 NON-FUNCTIONAL REQUIREMENTS:

- **Safety Requirements:**

As the image is between the sender and receiver both the sender and receiver should use same program to Encrypt and Decrypt. Sender and receiver should take care of no one should steal the code.

- **Security Requirements:**

We have developed a GUI software in which embedded secret text data. Only sender and receiver should be aware of encrypted file.

- **Software Quality Attributes:**

The Quality of the software is maintained in such a way that only sender and receiver can communicate through secret key.

3.3.3 SOFTWARE REQUIREMENTS:

3.3.3.1 Operating system: Windows 11.

3.3.3.2 Implementation: Visual Studio Code.

3.3.3.3 Packages: Tkinter, Base64, Random,Date Time etc.…,

3.3.4 HARDWARE REQUIREMENTS:

3.3.4.1 Minimum hardware requirements:

3.3.4.1.1 System: Intel i3 Processor
3.3.4.1.2 Hard Disk : 512 GB
3.3.4.1.3 Monitor: 15””LED
3.3.4.1.4 Input devices: Keyboard, Mouse
3.3.4.1.5 Ram: 4 GB

CHAPTER 4

VIGENERE CIPHER

4.1 TYPES OF VIGENERE CIPHER:

4.1.1 AUTOKEY METHOD:

In the world of cryptography, the Auto key Cipher is a classic technique used to secure messages. It is a type of substitution cipher where each letter of the plain text message is replaced by a letter determined by a key letter and the previous cipher letter

Vigenere Cipher Encoder in Auto Key Method

first step in the auto key method is to decide on a priming key. Both sender and receiver have to agree on this key. The priming key is the single alphabet that is added to the beginning of messages to help make the key.

The sender encrypts the message starting by writing the first letter of the plain text on one line and the priming key under it. The rest of the plain text is written as is, shifted one place to the right.

Plaintext	I	N	T	E	L	L	I	P	A	A
Key	R	I	N	T	E	L	L	I	P	A

The steps involved are as follows:

- Write the plain text.
- Use the plain text and the key letter to select a row and a column in the Vigenere table.
- The first letter of the plain text is the first row and the key is the first column. For example, if the plain text is INTELLIPAAT.
- the key is R, then the first row will be the one that starts with I, and the column will be the one that starts with R.
- The first letter of the ciphertext will be the letter where the first row and column intersect. In the case of our example that will be the letter Z.
- Now, this process is continued till the entire plain text is turned into a ciphertext. For INTELLIPAAT,that will be ZVGXPWTXPAT.

Plaintext	I	N	T	E	L	L	I	P	A	A	T
Key	R	I	N	T	E	L	L	I	P	A	A
Cipher text	Z	V	G	X	P	W	T	X	P	A	T

Vigenere Cipher Decoder in Auto Key Method

The below-mentioned steps are followed to decipher the ciphertext:

The first letter is selected using the priming key.

The first letter of the ciphertext is located in this row.

The first letter of the plain text will be the letter where the first row and column intersect.

This process is followed until the entire ciphertext is deciphered.

Key	R	I	N	T	E	L	I	P	A	A
Cipher text	Z	V	G	X	P	W	T	X	P	A
Plaintext	I	N	T	E	L	L	I	P	A	A

4.1.2 Keyword Method:

The Keyword Method also uses the Vigenere Cipher table to encrypt and decrypt the plain text. In this method, the key is usually more than one letter.

Vigenere Cipher Encoder in Keyword Method

The Vigenere Cipher encoder is similar to the auto key method. The difference is that instead of using just one priming key, the Vigenere Cipher encoder uses a keyword that is more than one letter.

So, there are endless possibilities of a possible keyword. The steps involved are as follows: The sender will write the keyword repeatedly under the plain text to form the key.

This will then be encoded, just like the auto key method, until an entire ciphertext is formed. For this example, let us use the phrase ALL IS WELL and the keyword as CAKE.

Plaintext	A	L	L	I	S	W	E	L	L
Key	C	A	K	E	C	A	K	E	C
Cipher text	C	L	V	M	U	W	P	P	N

Vigenere Cipher Decoder in Keyword Method

To decode this, the following steps can be followed:

Start by writing the keyword repeatedly till the length of the ciphertext.

Write the ciphertext under it and decode it just as is done in the auto-key method.

Key	C	A	K	E	C	A	K	E	C
Cipher text	C	L	V	M	U	W	P	P	N
Plaintext	A	L	L	I	S	W	E	L	L

Security

This method is much more secure than the auto key method, but that does not mean it is not vulnerable. The cipher is more secure with longer keywords.

4.1.3 Vigenere Cipher Calculator:

If the Vigenere table is not available or accessible, for whatever reason, then encryption and decryption processes can be carried out using the Vigenere algebraic formula. For this method to work, the alphabets have to be converted into numbers (0-25).

The encryption formula is:

Ei = (Pi + Ki) mod 26

The decryption formula is:

Di = (Ei – Ki) mod 26

Where E is the encipher, D is the decipher, P is the plaintext, and K is the key.

If the value of Di is ever negative, 26 is added to that value.

For example, let us take our plain text to be CYBER SECURITY.

So,

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

The key can be BEST. Encryption:

Ei = (Pi + Ki) mod 26

Plaintext	C	Y	B	E	R	S	E	C	U	R	I	T	Y
Plaintext value (P)	02	24	01	04	17	18	04	02	20	17	08	19	24
Key	B	E	S	T	B	E	S	T	B	E	S	T	B
Key value (K)	01	04	18	19	01	04	18	19	01	04	18	19	01
Ciphertext value (E)	03	02	19	23	18	22	22	21	21	21	00	12	25
Ciphertext	D	C	T	X	S	W	W	V	V	V	A	M	Z

Decryption: Di = (Ei – Ki) mod 26

Remember, if the value of Di is ever negative, 26 is added to that value.

Ciphertext	D	C	T	X	S	W	W	V	V	V	A	M	Z
Ciphertext value (E)	03	02	19	23	18	22	22	21	21	21	00	12	25
Key	B	E	S	T	B	E	S	T	B	E	S	T	B
Key value (K)	01	04	18	19	01	04	18	19	01	04	18	19	01
Plaintext value (P)	02	24	01	04	17	18	04	02	20	17	08	19	24
Plaintext	C	Y	B	E	R	S	E	C	U	R	I	T	Y

The keyword is repeated to match the length of the plaintext message. Each letter in the keyword is then used to determine the shift value for the corresponding letter in the plaintext.

However, it is used in many stronger encryption algorithms like the Advance Encryption Standard (AES) . Vigenere algorithm can be viewed algebraically if the letters A–Z are taken to be the numbers 0–25, Vigenere encryption formula is:

$Ci = (Pi + Ki) \bmod m$

Where,

C = Cipher text character

P = Plain text character

K = Key phrase character

m = Length of alphabets (i.e, 26 in Vigenere cipher)

CHAPTER 5
DESIGN METHODOLOGIES

5.1 VIGENERE CIPHER:

The Vigenere Cipher, invented by the French cryptographer Blaise de Vigenere in the 16th century, is a method of encrypting and decryption messages. It is a poly alphabetic substitution cipher, which means that it uses multiple cipher alphabets to encrypt the plain text.

The Vigenère Cipher operates by using a keyword, typically a word or phrase, as the basis for encryption. The keyword is repeated to match the length of the plain text message. Each letter in the keyword is then used to determine the shift value for the corresponding letter in the plain text.

To encrypt a message, each letter of the plain text is shifted by the corresponding letter in the keyword. This is done using a table called the Vigenère square or Vigenère tableau.

The Vigenère square is a grid of alphabets, where each row represents a shift of the previous row by one position. The letter at the intersection of the keyword letter and plain text letter in the Vigenère square gives the encrypted letter.

For example

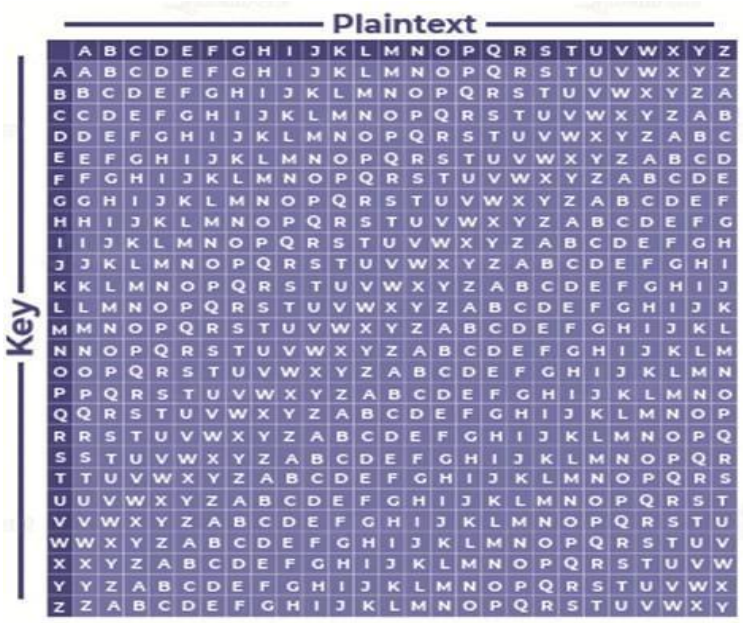
let's encrypt the word "HELLO" using the keyword "KEY". First, repeat the keyword to match the length of the plain text: "KEYKE". Then, find the corresponding letters in the Vigenère square. The letter 'H' is shifted by 'K', resulting in 'S'. 'E' is shifted by 'E', resulting in 'X'. 'L' is shifted by 'Y', resulting in 'M'. And so on. The encrypted message would be "SXMMR".

Decryption works in a similar way. Each letter of the cipher text is shifted back by the corresponding letter in the keyword, using the Vigenère square. By reversing the process, the original plain text message can be obtained.

The Vigenère Cipher offers more security compared to simple substitution ciphers because it introduces variability and makes it harder to identify patterns in the encrypted message. However, it is not as strong as modern encryption methods and can be vulnerable to cryptanalysis techniques such as frequency analysis and Kasiski examination.

5.1.1 Vigenere Cipher Table:

It table consists of all the alphabets written 26 times in different rows. Each alphabet in every subsequent row and column is shifted cyclically to the left. This generates 26 Caesar Ciphers. Different alphabets from different rows are used at different points in the encryption process. The alphabet chosen depends on the method used for encryption.



The Vigenere cipher is a method of encrypting alphabetic text by using a series of different Caesar ciphers based on the letters of a keyword. It is a simple form of poly alphabetic substitution in which each alphabet can replace with several cipher alphabets.

Vigenere cipher was considered secure for centuries but later its weakness was identified. Friedrich Kasiski discovered a method to identify the period and hence key and plain text .

The basic theme of Vigenere cipher is to conceal Plaintext letter frequencies by defeating simple frequency analysis. But the crucial weakness of the Vigenere cipher is the repeating nature of its key.

If a cryptanalyst correctly guesses the key's length, then the cipher text can be easily broken. Various methods like Kasiski and Friedman tests can help to determine the
key length.

With the advent of computer the Vigenere cipher has become even easier to break. Most cipher texts can be cracked within a few seconds even with long keys.

This cipher is now considered trivial to break and provides no security by today's standards. However, it is used in many stronger encryption algorithms like the Advance Encryption Standard (AES) . Vigenere algorithm can be viewed algebraically if the letters A–Z are taken to be the numbers 0–25, Vigenere encryption formula is:

$Ci = (Pi + Ki) \bmod m$

Where,

C = Cipher text character

P = Plain text character

K = Key phrase character

m = Length of alphabets (i.e, 26 in Vigenere cipher)

5.2 How easy or difficult is it to break the Vigenère cipher?

It should be noted here that we can employ a variety of techniques in order to decrypt a Caesar cipher and we don't even need to know about the Caesar cipher key.

That is because the patterns in the plain text message get preserved by the Caesar cipher encryption scheme. However, when compared with the Caesar cipher, it is more complex to break the Vigenère cipher.

In the Vigenère cipher, the number of places a letter in the plain text message is shifted in the alphabet changes for each letter in the plain text message

.

This, in turn, means that the most frequently used letters in the ciphertext do not correspond to the most frequently used letters in the plain text. Moreover, if we find a letter used more than once consecutively in the ciphertext, that doesn't mean that the consecutively used letter in the ciphertext correspond to a consecutively used letter in those positions in the plain text.

As a result of these features, it took a lot longer to break the Vigenère cipher than it took to break the Caesar cipher.However, having said all these, the Vigenère cipher is not unbreakable at all.

For example,

spies used the lines of famous poems as their Vigenère cipher keys in the Second World War. German codebreakers had figured out the pattern and it then became easier for them to guess a ciphertext's encryption key and eventually decrypt the message.

It is a matter of common sense that being able to guess the key that was used to encrypt a plaintext message will mean that the ciphertext message can be decrypted back to the original plaintext message.

Patterns in the ciphertext can still provide some clues. It is very likely that a sequence of letters repeated in multiple places in the ciphertext message could be the same word (or part of a word) of the plaintext message encrypted using the same part of the key.

It is quite less likely (could happen, but very less likelihood of happening) that such repetitions in the ciphertext message are different sets of letters in the plaintext message that were encrypted as the same pattern in the ciphertext message.

Codebreakers could use a technique called 'dragging' to crack the Vigenère cipher. Suppose that the codebreaker thinks that he/she may know about a word or phrase that will be in the plaintext message. The codebreaker can then test to see if his/her guessed word or phrase has been encrypted anywhere in the message until the time that such a process produces a key that appears not to be random.

When the codebreaker has a clue to the key, he/she can then try decrypting the rest of the ciphertext message. By doing that, the codebreaker will have an idea if he/she has been able to discover the whole key - if not, the codebreaker has to try multiple approaches which is usual with the 'dragging' process, until enough patterns in the message, that can help in decrypting the ciphertext message back to the original plaintext message, can be spotted.

5.3 Modified Vigenere Cipher Algorithm in our Project:

For modification of Vigenère cipher, numbers, punctuations, mathematical symbols may be used for key in place of characters. Random generated numbers may be used to spread the encrypted message. both encryption and decryption process are carried out using a single key**.**

modified version of vigenere algorithm was proposed in which diffusion is provided by adding a random bit to each byte before the message is encrypted using Vigenere.

This technique fails kasiski attack to find the length of key because the padding of message with random bits. The main drawback of this technique is that the size of the
encrypted message will be increased by around 56%.

In a new way of implementing Vigenere algorithm was introduced by automatically changing the cipher key after each encryption step. In this method successive keys were used that were dependent on the initial key value during the encryption process.

In modification of Vigenere cipher by random numbers, punctuations & mathematical symbols was presented. In proposed method numbers, punctuations and mathematical symbols were used for key in place of characters to make it more difficult for brute force attack.

It was concluded that if random numbers are used for key and to spread the spectrum then only skilled persons can identify the message.

A novel approach was presented by combing the LFSR (Linear feedback shift register) key
with Vigenere cipher key. In Proposed technique the concepts and methodologies of classical Vigenere cipher plus modern LFSR stream cipher was used.

Proposed technique generates the period less and pseudorandom letter key stream for encryption/ decryption. Period less and randomness of key lowers the flatness of letter frequency that makes it difficult to identify the length of key.

In the Caesar Cipher and Vigenere Cipher have been modified and expanded by including alphabets, numbers and symbols and at the same time introduced a complete confusion and diffusion into the modified cipher developed.

It was concluded that cipher text generated by proposed hybrid technique is very difficult to break using a frequency method, brute force attack etc.

A new algorithm by combining Vigenere substitution cipher with Stream cipher was proposed in which repeated portions of plaintext always encrypted with the different portion of the keyword or binary key.

The letters in odd location were encrypted with stream cipher and the letters in even locations with Vigenere cipher.

It was concluded that proposed algorithm hides the relation between cipher text and plain text that makes cryptanalysis much difficult.

5.4 Why we use the Vigenere Cipher Algorithm in Encrypting and Decrypting the data:

The Vigenère cipher is a method of encrypting messages by using a series of different Caesar ciphers based on the letters of a particular keyword. The Vigenère cipher is more powerful than a single Caesar cipher and is much harder to crack.

vigenere cipher algorithm method is used for securing data in the form of digital images, especially for the logo of a company that is one of the methods of science of cryptography, in which the order logos or digital image that secure during transmission over the Internet.

The Vigenère cipher is based upon a square that consists of the 26 Caesar cipher alphabets; this is in fact the square.The key to this method of encryption is a memorable word or phrase.

The Vigenère Cipher is a simple form of poly alphabetic substitution through which alphabetic text is encrypted using a series of Caesar ciphers with different shift values based on the letters of a keyword.

So much flexible and now that Vigenere Cipher Algorithm became the new version.So by using this vigenere Cipher our Project became the more easy to perform encryption and decryption of data.In our project we only performing the encryption and decryption to only text not for images,.

CHAPTER 6

UML DIAGRAMS

UML which stands for Unified Modeling Language, is a way to visually represent the architecture, design, and implementation of complex software systems.

UML is an acronym that stands for Unified Modeling Language. Simply put, UML is a modern approach to modeling and documenting software. In fact, it's one of the most popular business process modeling techniques.

It is based on diagrammatic representations of software components. As the old proverb says: "a picture is worth a thousand words". By using visual representations, we are able to better understand possible flaws or errors in software or business processes

The UML diagram, grounded in the Unified Modeling Language (UML), is defined as a dynamic template facilitating the visualization of processes and sequences. This visual representation meticulously captures a system's essential elements, including actors, roles, actions, and artifacts.

With a primary goal of not only enhancing understanding but also facilitating seamless alterations, maintenance, and comprehensive documentation of crucial system information, UML diagrams stand as an indispensable tool in modern software design and development.

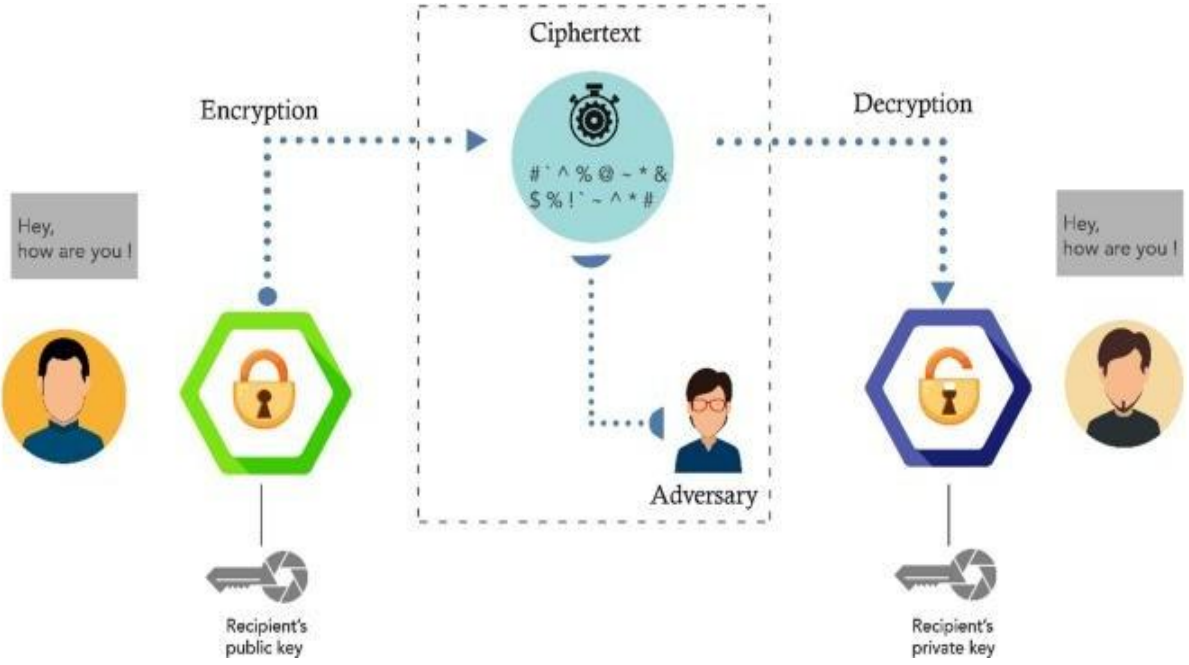
UML was not always a thing, but its creation resulted from the chaos around software development and documentation in the late 1990s. There were many ways to describe and represent existing software systems. Because of this confusion, there was a need to develop a better way of visualizing those systems.

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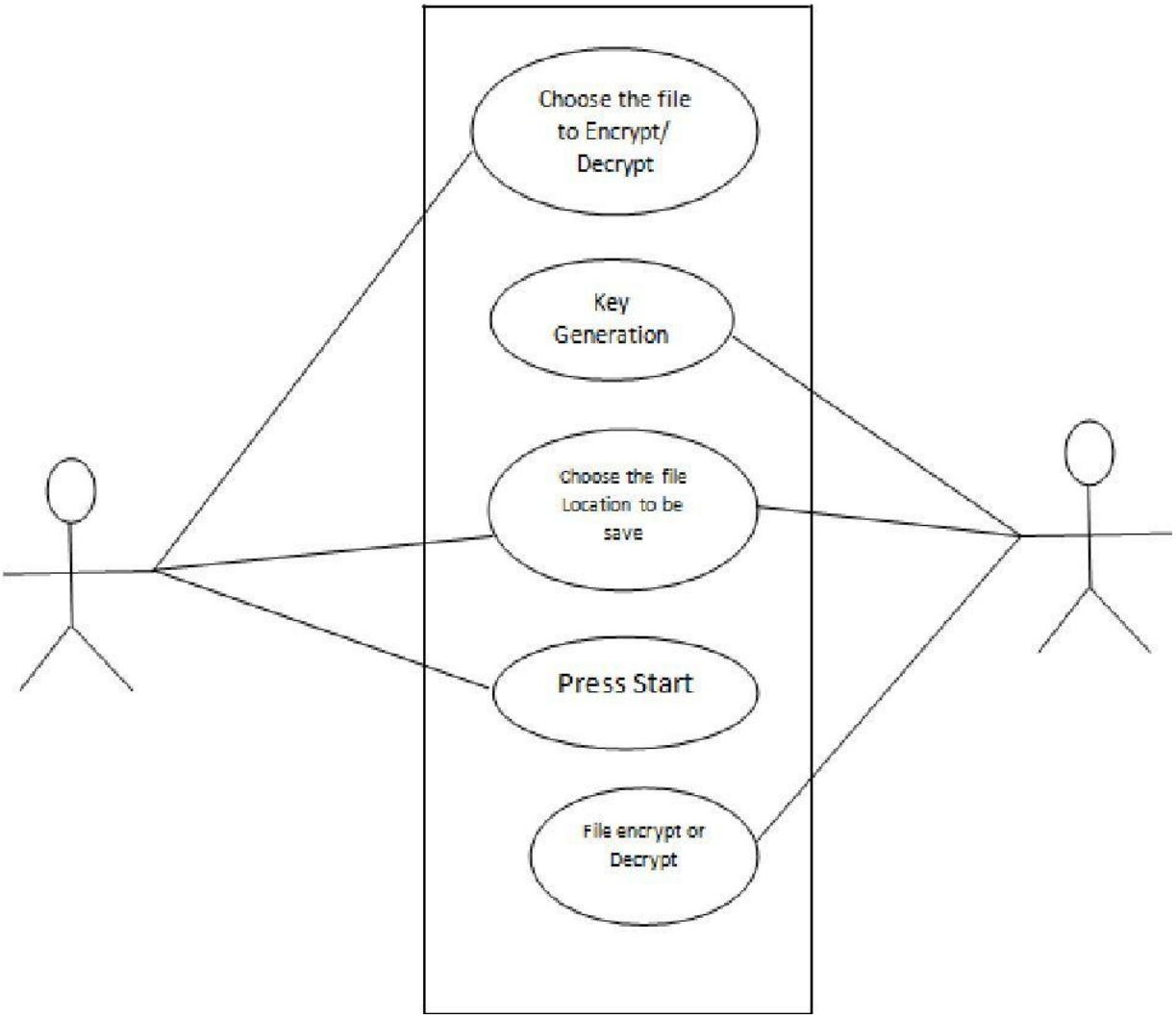
TYPES OF UML DIAGRAMS:

- Class Diagram
- Use case Diagram
- Sequence Diagram
- Data Flow Diagram

6.1 DATAFLOW DIAGRAM:

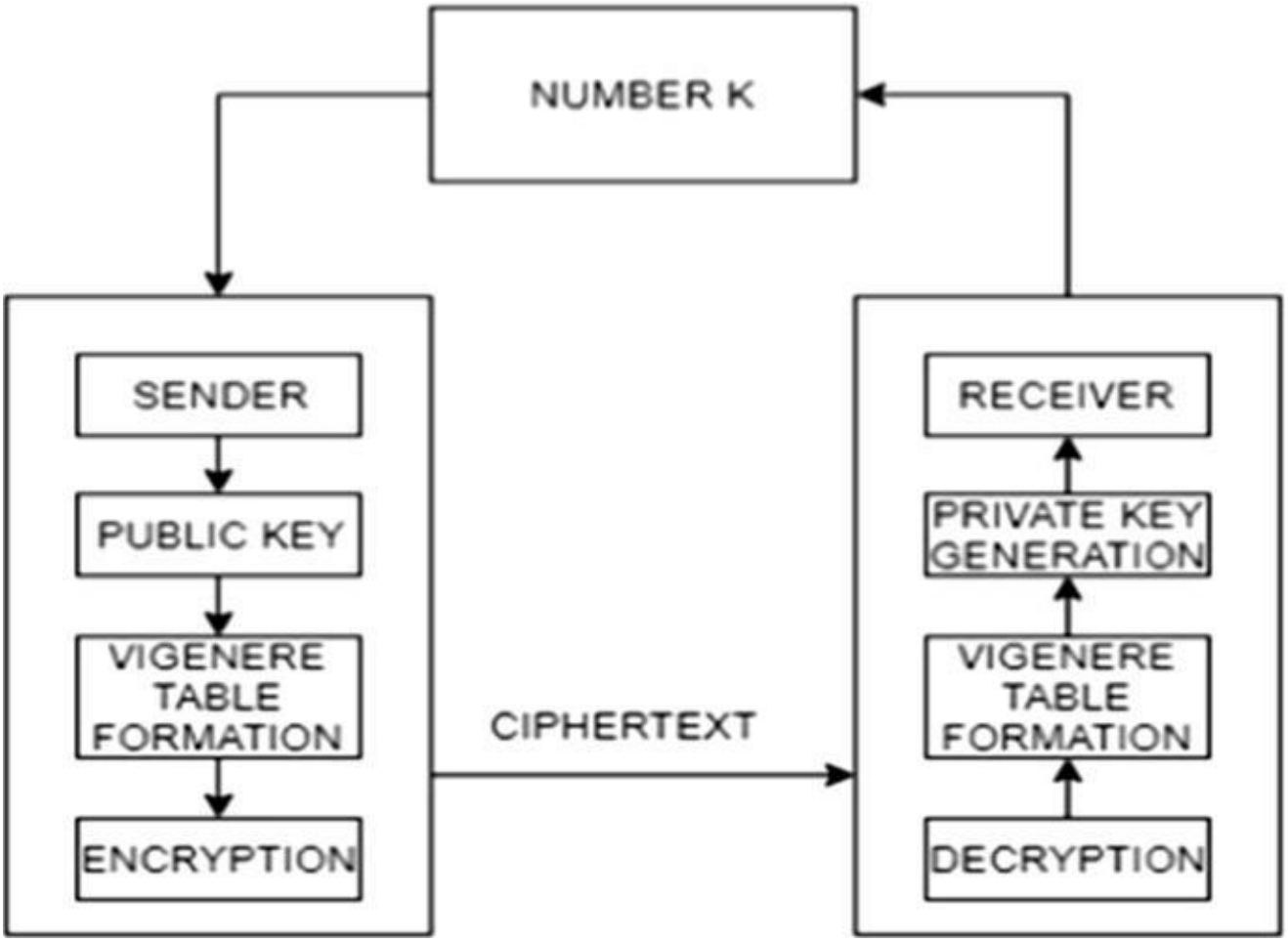


6.2 USE CASE DIAGRAM:



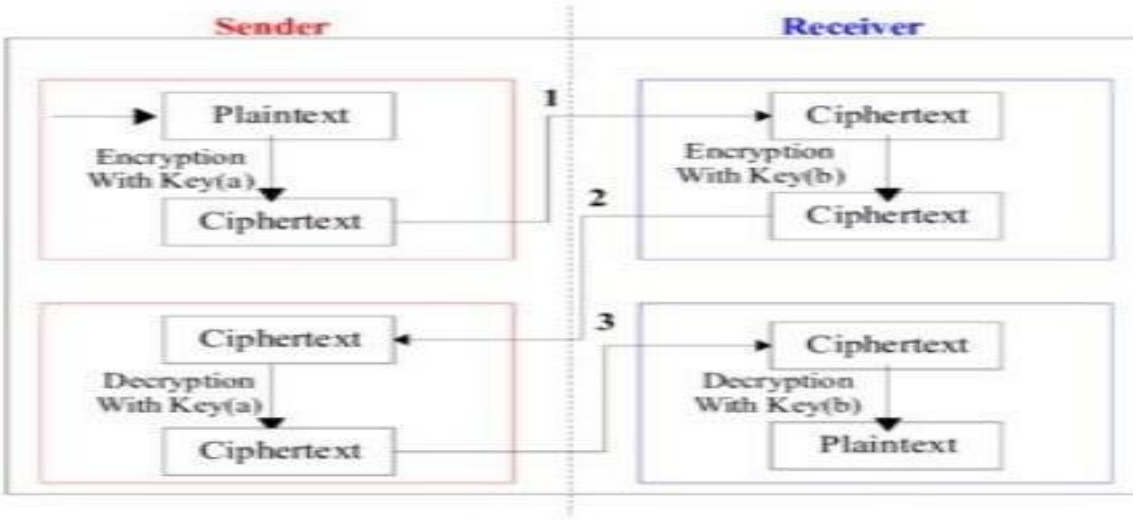
- use case diagram at its simplest is a representation of a user's interaction with the system and depicting the specifications of a use case.
- A use case diagram can portray the different types of users of a system and the various ways that they interact the system.
- This type of diagram is typically used in conjunction with the textual use case and will often be accompanied by other types of diagrams as well.

6.3 CLASS DIAGRAM:



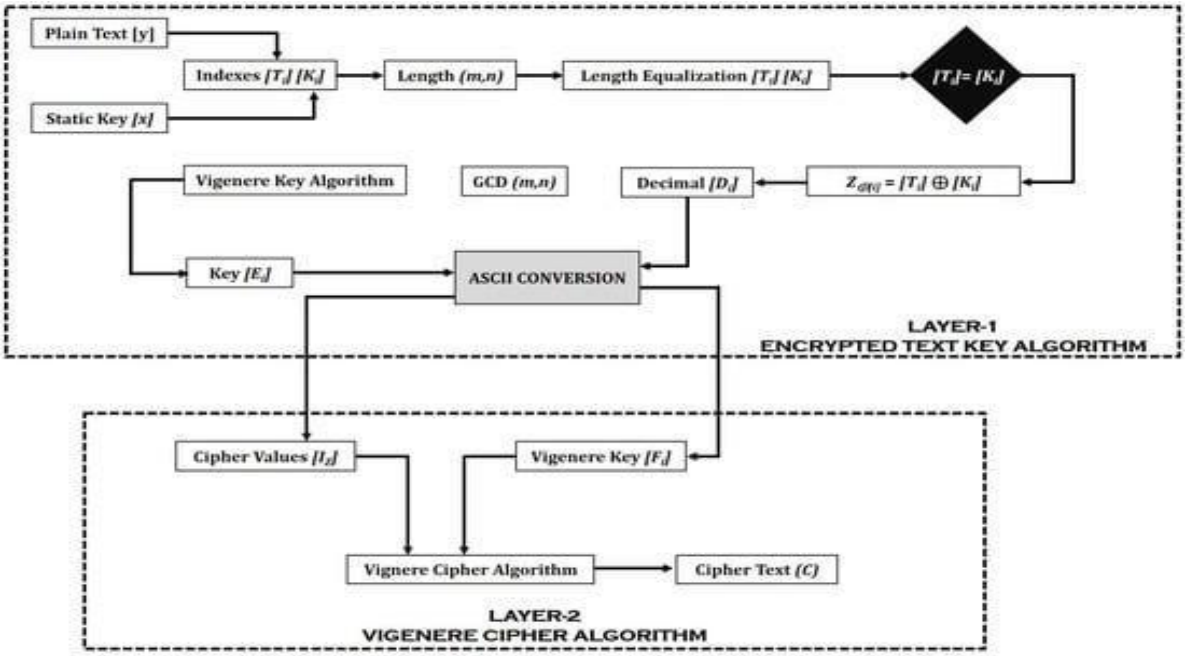
- In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among objects.

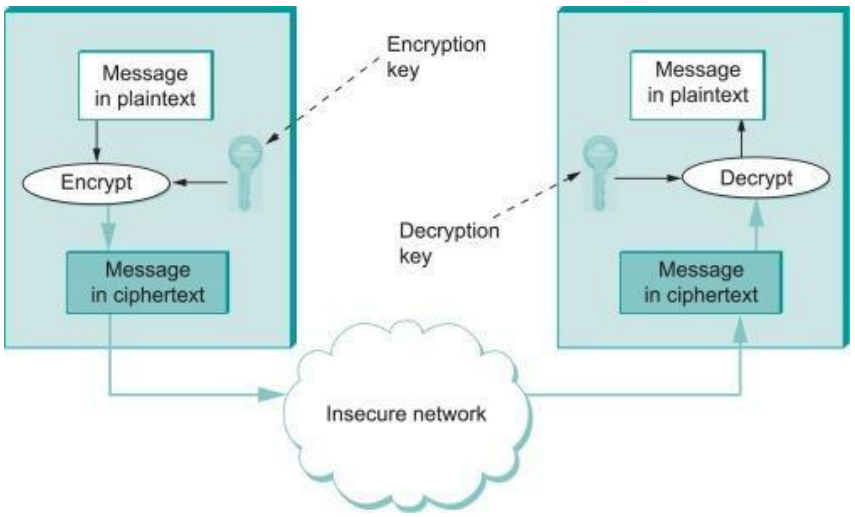
6.4 SEQUENCE DIAGRAM:



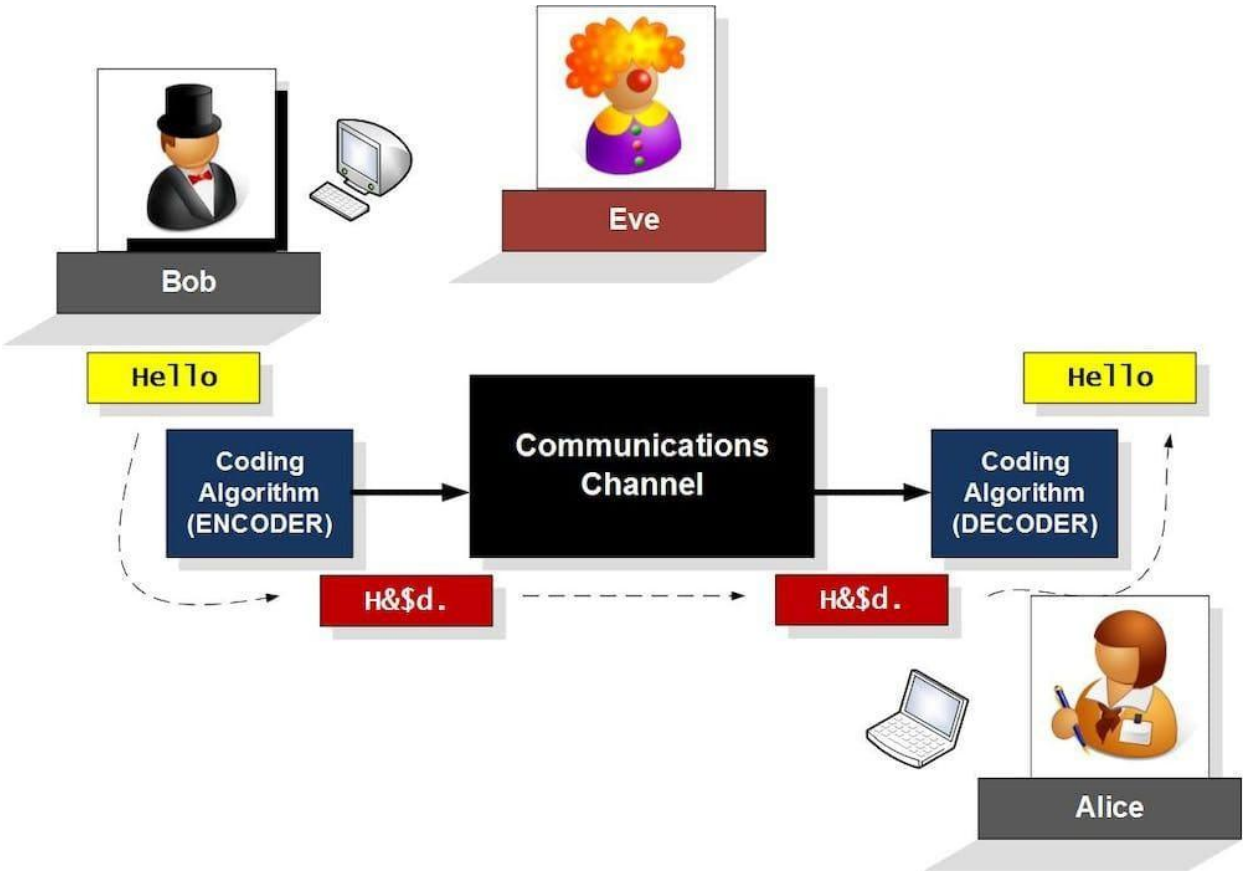
- Unified Modelling Language (UML) is a modeling language in the field of software engineering which aims to set standard ways to visualize the design of a system.
- UML guides the creation of multiple types of diagrams such as interaction , structure and behaviour diagrams. A sequence diagram is the most commonly used interaction diagram.

6.5 TWO LAYER SECURITY ALGORITHM:





6.6 Activity Diagram:



- An activity diagram visually presents a series of actions or flow of control in a system similar to a flowchart or a data flow diagram.
- Activity diagrams are often used in business process modeling. They can also describe the steps in a use case diagram.

6.7 SOURCE CODE:

Here we use visual Studio Code for doing this code.

```
# import Tkinter module
from Tkinter import *

# import other necessary modules
import random
import time
import date time

# creating root object
root = Tk()

# defining size of window
root.geometry("1200x6000")
# setting up the title of window
root.title("Message Encryption and Decryption")

Tops = Frame(root, width = 1600, relief = SUNKEN)
Tops.pack(side = TOP)

f1 = Frame(root, width = 800, height = 700,
                        relief = SUNKEN)
f1.pack(side = LEFT)

# ==============================================
#                 TIME
# ==============================================
local time = time.asctime(time.local time(time.time()))

lblInfo = Label(Tops, font = ('helvetica', 50, 'bold'),
        text = "SECRET MESSAGING \n Vigenère cipher",
```

```
                fg = "Black", bd = 10, anchor='w')

lblInfo.grid(row = 0, column = 0)

lblInfo = Label(Tops, font=('arial', 20, 'bold'),
          text = localtime, fg = "Steel Blue",
                    bd = 10, anchor = 'w')

lblInfo.grid(row = 1, column = 0)

rand = StringVar()
Msg = StringVar()
key = StringVar()
mode = StringVar()
Result = StringVar()

# exit function
def qExit():
    root.destroy()

# Function to reset the window
def Reset():
    rand.set("")
    Msg.set("")
    key.set("")
    mode.set("")
    Result.set("")



# reference
lblReference = Label(f1, font = ('arial', 16, 'bold'),
               text = "Name:", bd = 16, anchor = "w")

lblReference.grid(row = 0, column = 0)

txtReference = Entry(f1, font = ('arial', 16, 'bold'),
```

```
            textvariable = rand, bd = 10, insertwidth = 4,
                    bg = "powder blue", justify = 'right')

txtReference.grid(row = 0, column = 1)

# labels
lblMsg = Label(f1, font = ('arial', 16, 'bold'),
        text = "MESSAGE", bd = 16, anchor = "w")

lblMsg.grid(row = 1, column = 0)

txtMsg = Entry(f1, font = ('arial', 16, 'bold'),
        textvariable = Msg, bd = 10, insertwidth = 4,
            bg = "powder blue", justify = 'right')

txtMsg.grid(row = 1, column = 1)

lblkey = Label(f1, font = ('arial', 16, 'bold'),
          text = "KEY", bd = 16, anchor = "w")

lblkey.grid(row = 2, column = 0)

txtkey = Entry(f1, font = ('arial', 16, 'bold'),
        textvariable = key, bd = 10, insertwidth = 4,
            bg = "powder blue", justify = 'right')

txtkey.grid(row = 2, column = 1)

lblmode = Label(f1, font = ('arial', 16, 'bold'),
         text = "MODE(e for encrypt, d for decrypt)",
                        bd = 16, anchor = "w")

lblmode.grid(row = 3, column = 0)

txtmode = Entry(f1, font = ('arial', 16, 'bold'),
        textvariable = mode, bd = 10, insertwidth = 4,
```

```
            bg = "powder blue", justify = 'right')

txtmode.grid(row = 3, column = 1)

lblService = Label(f1, font = ('arial', 16, 'bold'),
        text = "The Result-", bd = 16, anchor = "w")

lblService.grid(row = 2, column = 2)

txtService = Entry(f1, font = ('arial', 16, 'bold'),
        textvariable = Result, bd = 10, insertwidth = 4,
                bg = "powder blue", justify = 'right')

txtService.grid(row = 2, column = 3)

# Vigenère cipher
import base64

# Function to encode
def encode(key, clear):
    enc = []

    for i in range(len(clear)):
        key_c = key[i % len(key)]
        enc_c = chr((ord(clear[i]) +
                ord(key_c)) % 256)

        enc.append(enc_c)
    return base64.urlsafe_b64encode("".join(enc).encode()).decode()

# Function to decode
def decode(key, enc):
    dec = []

    enc = base64.urlsafe_b64decode(enc).decode()
    for i in range(len(enc)):
```

```
        key_c = key[i % len(key)]
        dec_c = chr((256 + ord(enc[i]) -
                     ord(key_c)) % 256)

        dec.append(dec_c)
    return "".join(dec)

def Ref():
    print("Message= ", (Msg.get()))

    clear = Msg.get()
    k = key.get()
    m = mode.get()

    if (m == 'e'):
        Result.set(encode(k, clear))
    else:
        Result.set(decode(k, clear))

# Show message button
btnTotal = Button(f1, padx = 16, pady = 8, bd = 16, fg = "black",
                  font = ('arial', 16, 'bold'), width = 10,
                  text = "Show Message", bg = "powder blue",
                   command = Ref).grid(row = 7, column = 1)

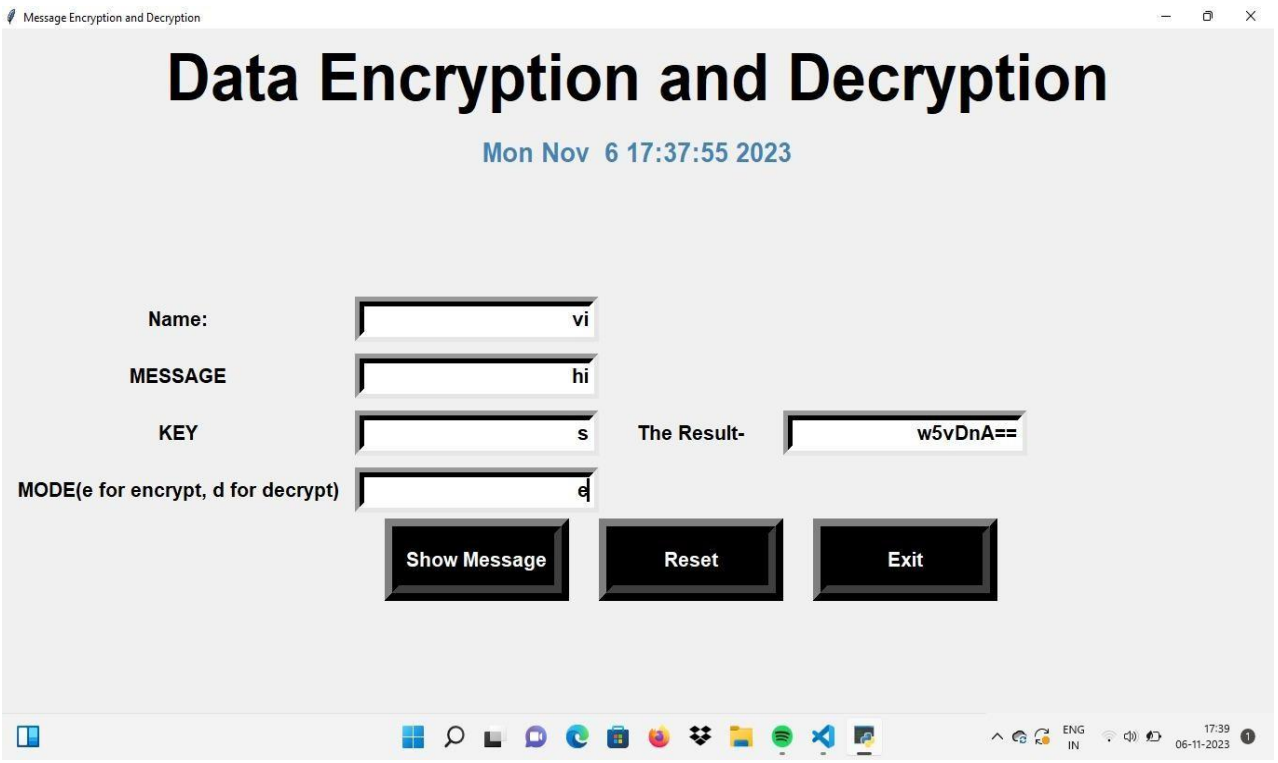
# Reset button
btnReset = Button(f1, padx = 16, pady = 8, bd = 16,
                fg = "black", font = ('arial', 16, 'bold'),
                width = 10, text = "Reset", bg = "green",
                 command = Reset).grid(row = 7, column = 2)

# Exit button
btnExit = Button(f1, padx = 16, pady = 8, bd = 16,
                fg = "black", font = ('arial', 16, 'bold'),
                    width = 10, text = "Exit", bg = "red",
                command = qExit).grid(row = 7, column = 3)
```

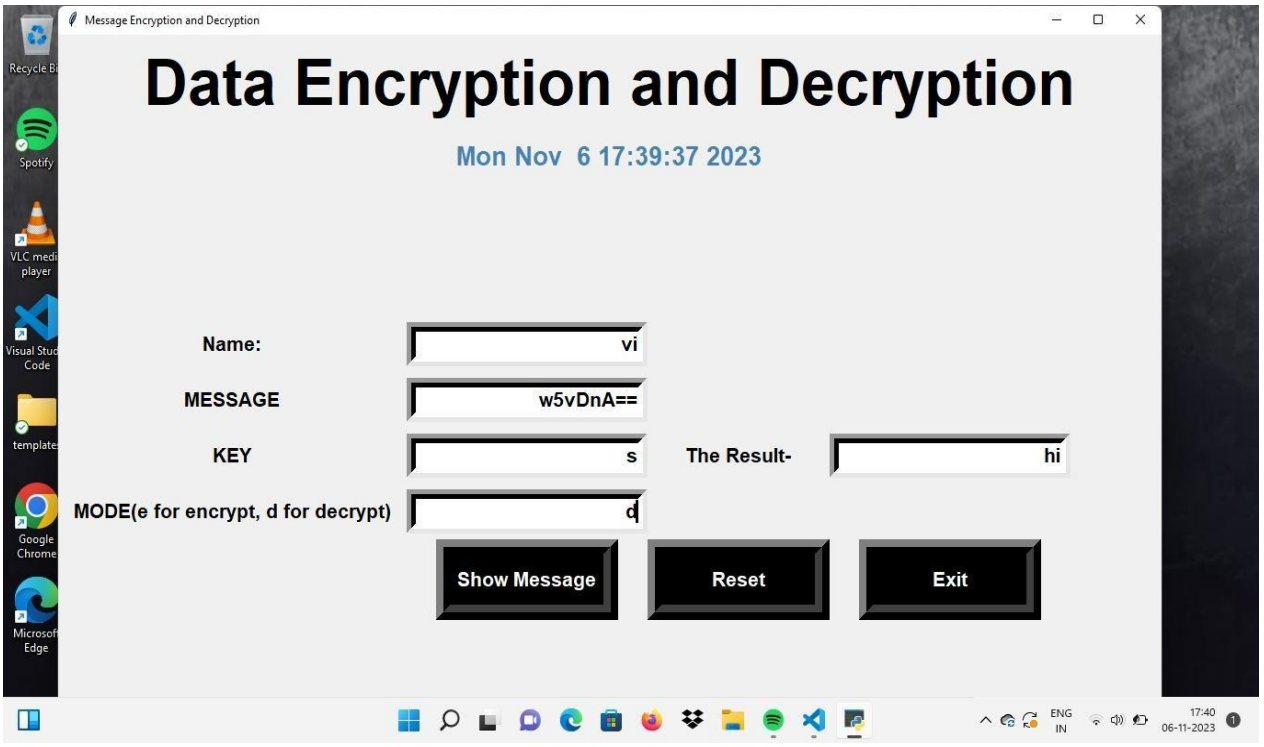
root.mainloop()

6.7 OUTPUT OF THE PROJECT:

6.7.1 ENCODE:



6.7.2 DECODE:


Message Encryption and Decryption
Data Encryption and Decryption
Mon Nov 6 17:39:37 2023
Name:
vi
MESSAGE
w5vDnA==
KEY
s
The Result-
hi
MODE(e for encrypt, d for decrypt)
d
Show Message
Reset
Exit

CHAPTER 7

APPLICATIONS OF SYSTEM

7.1 Advantages of Vigenère Cipher:

- **Poly alphabetic Encryption:** The Vigenère Cipher uses multiple substitution alphabets based on a keyword, making it more complex than simple substitution ciphers. This complexity can make it harder for frequency analysis attacks to be effective.
- **Resistance to Simple Frequency Analysis:** The Vigenère Cipher resists basic frequency analysis because each letter can be encoded in multiple ways, depending on its position in the keyword.
- **Relatively Easy to Learn:** The Vigenère Cipher is straightforward to understand and implement, making it a good introduction to the concept of poly alphabetic ciphers.
- **Key Management:** The Vigenère cipher relies on a keyword or phrase as the encryption key. This key can be easily changed for different messages, providing flexibility in encryption. The key length and complexity can enhance the security of the cipher.
- **Vigenère Square:** A Vigenère square is a grid or matrix formed by writing the alphabets repeatedly, starting at different places. The following figure depicts a Vigenère square.
- Same letter can be be encoded in n different ways.
- Use each alphabet works in turn.
- Effectively multiple Caesar ciphers.
- Difficult to detect.
- Only receiver can detect.
- It can be done faster with large number of software's .
- Can be applied differently in digital image, audio and video files.
- It is used in way of hiding not the information but the password to reach that information.
- Can provide digital signatures that can be repetitive.

7.2 APPLICATIONS OF VIGENERE CIPHER:

1. Secure Private Files and Documents.
2. Hide Passwords and Encryption Keys.
3. Transport Highly Private Documents between International Governments.
4. Transmit message and data without revealing the existence of available message.
5. Confederate cipher disk that was used as a field cipher during the American Civil War.
6. Confidential communication and secret data storing.
7. Protection of data alteration.
8. Access control system for digital content distribution.
9. Media Database systems.
10. Defense and intelligence.
11. Medical.
12. Online Banking.
13. Other financial and commercial purposes.
14. Vigenère cipher method is used for securing the data in the form of digital images, especially for the logo of a company that is one of the method of science of cryptography that secures during transmission over the internet.
15. Vigenère cipher can be used to enhance the security of various applications, including secure communication.
16. Different approaches to secure communication, as discussed in the next section, entail different implementations of Vigenère Cipher.

CHAPTER 8
RESULT & FUTURE WORK

8.1 RESULT:

After the completion of the project the users encrypt their important data and also encrypt their file. Data security is one of the advantage by using the encryption and decryption.

8.2 FUTURE WORK :

Cryptography is the widely used method for the security of data. Vigenère cipher is one of the cryptographic method that is considered simplest and weakest due to many limitations.

To overcome the limitations of Vigenère cipher we proposed an enhanced version of Vigenère cipher that is much secure against Kasiski and Friedman attacks. Cryptanalysis, frequency analysis, pattern prediction and brute attack on proposed technique are also much difficult due to use of multiples tables for encryption.

Although there are many cryptographic methods but this domain still requires serious attention of research community for the improvement of data security. In future our aim is to provide validation of proposed approach by performing security and performance analysis.

CHAPTER 9
CONCLUSION & REFERENCE

9.1 CONCLUSION:

Hence, this Vigenère cipher and cryptography method enables the conduct of business over the networks using the computer systems in extremely efficient and effective manner. And moreover, it ensures the safety of user's critical data.

Since it is a simple encryption method it is easier to use than other complex encryption methods moreover, it is secure enough to safeguard the sensitive data of the users in contrary to its simpler counterparts.

9.2 REFERENCE:

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